

Machine Learning-Based Battery Health Prediction and Preventive Maintenance Framework for Electric Vehicle Fleets

Overview

This dissertation develops a data-driven battery diagnostic and preventive maintenance decision framework for EV fleets using Machine Learning. The study applies multiple classification and degradation prediction models to open-source battery cycling data, compares their performance quantitatively, and builds a maintenance decision framework from the best-performing results. The work is fully simulation and data-based (Python and MATLAB). No physical lab work or human participants are involved.

Selected Datasets

- NASA Prognostics Battery Dataset — lithium-ion charge/discharge cycling data under controlled laboratory conditions.
- CALCE Battery Research Group Dataset — capacity fade and impedance measurements across cell chemistries and cycling conditions.
- Oxford Battery Degradation Dataset — long-term structured ageing experiments with high-fidelity degradation profiles.

ML Models — Comparative Evaluation

Six models will be trained, tuned, and benchmarked before final model selection. Selection will be evidence-based, documented in the methodology chapter.

Model	Category	Primary Use	Metrics
Random Forest	Ensemble	Fault classification	Accuracy, F1-score
XGBoost	Ensemble	Fault & SoH prediction	Accuracy, F1-score
Gradient Boosting	Ensemble	Regression / SoH ranking	RMSE, MAE
SVM	Kernel-based	Binary fault detection	Accuracy, F1-score
Logistic Regression	Linear	Baseline classifier	Accuracy, F1-score
LSTM	Deep Learning	Time-series degradation forecast	RMSE, MAE

Evaluation Metrics

Metric	Formula	Applied To
Accuracy	$\text{Correct predictions} / \text{Total predictions}$	Classification models (RF, XGBoost, SVM, LR)
F1-Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Imbalanced fault class detection
RMSE	$\sqrt{\sum(\text{predicted} - \text{actual})^2 / n}$	SoH regression — LSTM, Gradient Boosting
MAE	$\sum \text{predicted} - \text{actual} / n$	SoH regression — interpretability

Expected Outputs

- Trained ML model pipeline with comparative performance dashboard across all six models and four metrics.
- SoH degradation prediction curves with confidence intervals (LSTM); fault classification report with confusion matrices and feature importance.
- Fleet-level preventive maintenance decision framework with rule-based triggers from model outputs.
- Dissertation: ~12,500 words across 5 chapters. Target publication: Journal of Energy Storage or Applied Energy.

Completion Plan

Phase	Timeline	Key Activities
Setup & Lit Review	May 2026	Supervisor confirmation, EG7030 registration, literature review, dataset download, Python/MATLAB environment setup.
Data Preparation	Early June 2026	Exploratory data analysis, feature engineering, data cleaning and normalisation across all three datasets.
Model Development	Mid June – July 2026	Train and compare all 6 ML models (RF, XGBoost, GB, SVM, LR, LSTM). Hyperparameter tuning, cross-validation, evidence-based model selection.
Framework Design	Late July 2026	Preventive maintenance decision framework built from model outputs. Powertrain context documented at conceptual level.
Results & Analysis	Early August 2026	Results interpreted, visualisations prepared, discussion and limitations drafted.
Writing & Submission	Aug – Early Sept 2026	Full dissertation written (~12,500 words), supervisor review completed, submitted via UEL portal.